Stochastic Processes 2021/22 Semester 1

Extension Questions 5

These questions are an extension to Tutorial 5, intended for personal revision only. Answers are available for all questions, but worked solutions are only available for some.

1. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5,6,7\}$ and has transition matrix

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1/8 & 1/3 & 1/12 & 0 & 1/3 & 1/8 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 4/5 & 1/5 & 0 & 0 & 0 \\ 0 & 1/8 & 1/8 & 1/8 & 0 & 1/8 & 1/2 \\ 1/5 & 0 & 0 & 0 & 0 & 4/5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

   (a) Identify the closed classes.

   (b) Determine the limiting distribution if the Markov chain starts from the initial state $X_0 = 1$.

   (c) Determine the probability of absorption into each of the closed classes if the Markov chain starts from the state $X_0 = 2$.

   (d) Determine the limiting distribution if the Markov chain starts from $X_0 = 2$.

2. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5,6,7,8\}$ and has transition matrix

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 1/2 & 0 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2/3 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 1/3 & 0 & 0 & 2/3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Classify the state space.

(b) Does there exist a limiting distribution if the Markov chain starts

i. from the state $X_0 = 2$?

ii. from the state $X_0 = 5$?

iii. from a random initial value with distribution $\mu = (\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$?

iv. from a random initial value with distribution $\mu = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, 0, 0)$?

v. from a random initial value with distribution $\mu = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, 0, 0, 0)$?

vi. from a random initial value with distribution $\mu = (\frac{1}{8}, \frac{1}{3}, \frac{1}{4}, 0, \frac{1}{6}, \frac{1}{16}, \frac{1}{16}, 0)$?

vii. from a random initial value with distribution $\mu = (\frac{1}{8}, \frac{1}{4}, \frac{3}{16}, \frac{1}{16}, \frac{1}{6}, \frac{1}{16}, \frac{1}{16}, \frac{1}{12})$?

Provide your reasoning.

(c) Find the limiting distribution, assuming that the Markov chain starts from a case where one exists.

3. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1, 2, 3, 4, 5, 6, 7\}$ and has transition matrix

$$P = \begin{pmatrix} 0 & 2/3 & 1/3 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/6 & 1/18 & 1/9 & 1/3 & 2/9 & 0 & 1/9 \\ 1/12 & 0 & 1/12 & 1/6 & 1/3 & 2/9 & 1/9 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix}$$

(a) Identify the closed classes.

(b) Explain why there is no limiting distribution if the Markov chain starts from the initial state $X_0 = 1$

(c) Determine the probability of absorption into each of the closed classes if the Markov chain starts from the state $X_0 = 4$

(d) Find the probability that, starting from state $X_0 = 4$, the chain reaches the set $\{2, 3\}$ before it reaches $\{1\}$, given that it reaches the set $\{1, 2, 3\}$ at all.

(e) Explain why a limiting distribution exists if the chain starts in state $X_0 = 4$. Find it.